Data Engineering
101 - Kafka

KAFKA: Concepts

kafka

KAFKA BROKER

A KAFKA BROKER IS A SERVER THAT RUNS THE KAFKA SOFTWARE AND IS RESPONSIBLE FOR STORING AND SERVING DATA. BROKERS RECEIVE MESSAGES FROM PRODUCERS, ASSIGN OFFSETS TO MESSAGES, AND STORE THEM ON DISK.

EXAMPLE: IN A KAFKA CLUSTER, MULTIPLE BROKERS WORK TOGETHER TO ENSURE DATA IS RELIABLY STORED AND SERVED.

Shwetank Singh
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TOPICS

TOPICS ARE LOGICAL CHANNELS TO WHICH MESSAGES ARE SENT BY PRODUCERS AND FROM WHICH MESSAGES ARE READ BY CONSUMERS. A TOPIC IS DIVIDED INTO MULTIPLE PARTITIONS TO ALLOW PARALLEL PROCESSING.

EXAMPLE: A "USER_ACTIVITY" TOPIC MIGHT BE DIVIDED INTO SEVERAL PARTITIONS TO HANDLE HIGH MESSAGE VOLUME.

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PARTITIONS

PARTITIONS ARE SUBDIVISIONS OF TOPICS. EACH PARTITION IS AN ORDERED, IMMUTABLE SEQUENCE OF MESSAGES THAT IS CONTINUALLY APPENDED TO. PARTITIONS ENABLE KAFKA TO SCALE HORIZONTALLY AND MAINTAIN MESSAGE ORDER.

EXAMPLE: PARTITION 0 OF THE "USER_ACTIVITY" TOPIC STORES MESSAGES FOR A SPECIFIC SUBSET OF USERS.

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PRODUCERS

PRODUCERS ARE CLIENTS THAT SEND MESSAGES TO KAFKA TOPICS. THEY CAN SEND MESSAGES TO SPECIFIC PARTITIONS BASED ON A PARTITIONING STRATEGY OR DISTRIBUTE THEM EVENLY ACROSS ALL PARTITIONS.

EXAMPLE: A WEB APPLICATION THAT LOGS USER ACTIVITY SENDS THESE LOGS TO A KAFKA TOPIC AS MESSAGES.

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CONSUMERS

CONSUMERS ARE CLIENTS THAT READ MESSAGES FROM KAFKA TOPICS. CONSUMERS CAN OPERATE INDIVIDUALLY OR AS PART OF A CONSUMER GROUP, WHICH ALLOWS FOR PARALLEL PROCESSING OF MESSAGES.

EXAMPLE: AN ANALYTICS SERVICE READS USER ACTIVITY LOGS FROM A KAFKA TOPIC TO GENERATE REPORTS.

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CONSUMER GROUPS

CONSUMER GROUPS ALLOW MULTIPLE CONSUMERS TO COLLABORATE ON PROCESSING MESSAGES FROM A TOPIC. EACH PARTITION IN A TOPIC IS ASSIGNED TO ONLY ONE CONSUMER WITHIN A GROUP AT A TIME, ENSURING PARALLEL PROCESSING AND LOAD BALANCING.

EXAMPLE: THREE CONSUMERS IN A GROUP PROCESS MESSAGES FROM SIX PARTITIONS.

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OFFSETS

OFFSETS ARE UNIQUE IDENTIFIERS ASSIGNED TO EACH MESSAGE WITHIN A PARTITION. CONSUMERS USE OFFSETS TO TRACK WHICH MESSAGES HAVE BEEN READ.

EXAMPLE: A CONSUMER READS MESSAGES UP TO OFFSET 105 AND RESUMES FROM OFFSET 106 AFTER A RESTART.

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KAFKA CLUSTER

A KAFKA CLUSTER IS COMPOSED OF MULTIPLE BROKERS THAT WORK TOGETHER. CLUSTERS PROVIDE FAULT TOLERANCE AND HIGH AVAILABILITY.

EXAMPLE: A CLUSTER WITH THREE BROKERS CAN CONTINUE OPERATING IF ONE BROKER FAILS.

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REPLICATION

KAFKA REPLICATES PARTITIONS ACROSS MULTIPLE BROKERS TO ENSURE FAULT TOLERANCE. EACH PARTITION HAS A LEADER AND SEVERAL FOLLOWERS. THE LEADER HANDLES ALL READS AND WRITES, WHILE FOLLOWERS REPLICATE THE DATA.

EXAMPLE: PARTITION 0 HAS ONE LEADER AND TWO FOLLOWERS ACROSS THREE BROKERS.

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ZOOKEEPER

ZOOKEEPER IS USED FOR DISTRIBUTED COORDINATION AND METADATA MANAGEMENT IN KAFKA. IT MANAGES BROKER METADATA, LEADER ELECTION, AND CONFIGURATION.

EXAMPLE: ZOOKEEPER ENSURES A NEW LEADER IS ELECTED IF THE CURRENT LEADER BROKER FAILS.

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PRODUCERS AND ACKS

PRODUCERS SEND MESSAGES TO BROKERS AND CAN CONFIGURE ACKNOWLEDGMENT SETTINGS (ACKS) TO ENSURE RELIABLE MESSAGE DELIVERY.

EXAMPLE: A PRODUCER CONFIGURES ACKS TO WAIT FOR CONFIRMATION FROM ALL REPLICAS BEFORE CONSIDERING A MESSAGE SENT.

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RETENTION POLICY

KAFKA TOPICS CAN HAVE RETENTION POLICIES THAT DETERMINE HOW LONG MESSAGES ARE STORED. POLICIES CAN BE TIME-BASED OR SIZE-BASED.

EXAMPLE: A TOPIC IS CONFIGURED TO RETAIN MESSAGES FOR 7 DAYS, AFTER WHICH THEY ARE DELETED.

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LOG COMPACTION

LOG COMPACTION ENSURES THAT ONLY THE LATEST MESSAGE FOR EACH KEY IS RETAINED IN A TOPIC, USEFUL FOR MAINTAINING THE LATEST STATE. EXAMPLE: A LOG-COMPACTED TOPIC RETAINS ONLY THE LATEST UPDATE FOR EACH USER PROFILE.

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KAFKA CONNECT

KAFKA CONNECT IS A FRAMEWORK FOR INTEGRATING KAFKA WITH OTHER DATA SYSTEMS. IT PROVIDES CONNECTORS TO MOVE DATA IN AND OUT OF KAFKA.

EXAMPLE: USING KAFKA CONNECT TO SYNC DATA BETWEEN A MYSQL DATABASE AND A KAFKA TOPIC.

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KAFKA STREAMS

KAFKA STREAMS IS A LIBRARY FOR BUILDING STREAM PROCESSING APPLICATIONS ON TOP OF KAFKA. IT ALLOWS PROCESSING AND TRANSFORMING DATA IN REAL TIME.

EXAMPLE: AN APPLICATION USING KAFKA STREAMS AGGREGATES CLICKSTREAM DATA TO GENERATE REAL-TIME METRICS.

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MIRRORMAKER

MIRRORMAKER IS A TOOL FOR REPLICATING DATA BETWEEN KAFKA CLUSTERS, OFTEN USED FOR CROSS-DATACENTER REPLICATION.

EXAMPLE: USING MIRRORMAKER TO REPLICATE MESSAGES FROM A PRIMARY DATACENTER TO A BACKUP DATACENTER.

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KAFKA API

KAFKA PROVIDES APIS FOR PRODUCING, CONSUMING, AND MANAGING DATA, INCLUDING PRODUCER API, CONSUMER API, AND ADMIN API.

EXAMPLE: USING THE PRODUCER API TO SEND MESSAGES FROM A JAVA APPLICATION TO A KAFKA TOPIC.

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SECURITY

KAFKA SUPPORTS VARIOUS SECURITY FEATURES, INCLUDING SSL ENCRYPTION, SASL AUTHENTICATION, AND ACLS FOR AUTHORIZATION.

EXAMPLE: CONFIGURING SSL TO ENCRYPT DATA IN TRANSIT AND SASL FOR CLIENT AUTHENTICATION.

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ADMINCLIENT API

THE ADMINCLIENT API ALLOWS PROGRAMMATIC MANAGEMENT OF KAFKA TOPICS, BROKERS, AND CONFIGURATIONS.

EXAMPLE: USING ADMINCLIENT TO CREATE A NEW TOPIC AND CONFIGURE ITS RETENTION POLICY.

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MONITORING AND METRICS

KAFKA PROVIDES METRICS FOR MONITORING CLUSTER HEALTH AND PERFORMANCE. TOOLS LIKE PROMETHEUS AND GRAFANA CAN BE USED TO VISUALIZE THESE METRICS.

EXAMPLE: MONITORING CONSUMER LAG AND BROKER HEALTH USING PROMETHEUS AND GRAFANA DASHBOARDS.

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MESSAGE DELIVERY SEMANTICS

KAFKA SUPPORTS THREE TYPES OF MESSAGE DELIVERY SEMANTICS: AT MOST ONCE, AT LEAST ONCE, AND EXACTLY ONCE.

EXAMPLE: CONFIGURING A PRODUCER FOR EXACTLY-ONCE DELIVERY TO ENSURE NO MESSAGE IS LOST OR DUPLICATED.

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STATEFUL PROCESSING

KAFKA STREAMS SUPPORTS STATEFUL PROCESSING, ALLOWING APPLICATIONS TO MAINTAIN STATE ACROSS MESSAGES USING STATE STORES.

EXAMPLE: A STREAM PROCESSING APPLICATION THAT MAINTAINS A RUNNING COUNT OF EVENTS OVER A WINDOW OF TIME.

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WINDOWED OPERATIONS

KAFKA STREAMS PROVIDES SUPPORT FOR WINDOWED OPERATIONS, ENABLING TIME-BASED AGGREGATIONS AND TRANSFORMATIONS.

EXAMPLE: CALCULATING THE AVERAGE NUMBER OF USER CLICKS PER MINUTE USING WINDOWED OPERATIONS.

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KSQL

KSQL IS A SQL-LIKE INTERFACE FOR STREAM PROCESSING IN KAFKA, SIMPLIFYING THE CREATION OF STREAM PROCESSING APPLICATIONS.

EXAMPLE: USING KSQL TO FILTER, AGGREGATE, AND TRANSFORM STREAMS OF DATA IN REAL TIME.

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KAFKA ECOSYSTEM

KAFKA'S ECOSYSTEM INCLUDES VARIOUS TOOLS AND FRAMEWORKS FOR COMPREHENSIVE DATA PROCESSING, SUCH AS KAFKA CONNECT, KAFKA STREAMS, AND KSQL.

EXAMPLE: INTEGRATING KAFKA WITH A RELATIONAL DATABASE USING KAFKA CONNECT AND PROCESSING THE DATA WITH KAFKA STREAMS.

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PUBLISH/SUBSCRIBE MESSAGING

PUB/SUB SYSTEMS ALLOW DECOUPLING OF MESSAGE PRODUCERS AND CONSUMERS. KAFKA ACTS AS A BROKER FACILITATING THIS.

EXAMPLE: AN APPLICATION PUBLISHES USER ACTIVITY LOGS WHICH CAN BE CONSUMED BY ANALYTICS SERVICES.

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MESSAGE AND BATCHES

MESSAGES ARE THE BASIC UNIT OF DATA IN KAFKA, STORED AS BYTE ARRAYS. MESSAGES ARE WRITTEN IN BATCHES FOR EFFICIENCY.

EXAMPLE: A BATCH OF LOG MESSAGES SENT FROM AN APPLICATION.

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SCHEMAS

SCHEMAS DEFINE THE STRUCTURE OF MESSAGES, ENSURING CONSISTENCY. APACHE AVRO IS A COMMON SERIALIZATION FRAMEWORK USED WITH KAFKA.

EXAMPLE: AVRO SCHEMA FOR USER PROFILE DATA.

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TOPICS AND PARTITIONS

TOPICS ARE CATEGORIES TO WHICH MESSAGES ARE PUBLISHED. TOPICS ARE DIVIDED INTO PARTITIONS FOR SCALABILITY AND REDUNDANCY.

EXAMPLE: A "USER_ACTIVITY" TOPIC WITH PARTITIONS.

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PRODUCERS AND CONSUMERS

PRODUCERS CREATE AND SEND MESSAGES TO KAFKA TOPICS. CONSUMERS READ MESSAGES FROM TOPICS.

EXAMPLE: A MICROSERVICE PRODUCING ORDER DATA AND ANOTHER CONSUMING FOR PROCESSING.

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BROKERS AND CLUSTERS

A BROKER IS A KAFKA SERVER THAT STORES DATA AND SERVES CLIENTS. MULTIPLE BROKERS FORM A KAFKA CLUSTER, PROVIDING FAULT TOLERANCE AND SCALABILITY.

EXAMPLE: A KAFKA CLUSTER WITH THREE BROKERS.

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DISK-BASED RETENTION

KAFKA RETAINS MESSAGES ON DISK FOR A CONFIGURED PERIOD, ALLOWING CONSUMERS TO READ AT THEIR PACE.

EXAMPLE: RETAINING LOGS FOR 7 DAYS.

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MULTIPLE PRODUCERS AND CONSUMERS

KAFKA SUPPORTS MULTIPLE PRODUCERS AND CONSUMERS FOR THE SAME TOPIC, ENABLING FLEXIBLE DATA PIPELINES.

EXAMPLE: MULTIPLE SENSORS PRODUCING DATA TO A SINGLE TOPIC, MULTIPLE ANALYTICS SERVICES CONSUMING IT.

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HIGH THROUGHPUT

KAFKA CAN HANDLE LARGE VOLUMES OF MESSAGES EFFICIENTLY DUE TO ITS ARCHITECTURE.

EXAMPLE: PROCESSING MILLIONS OF LOG ENTRIES PER SECOND.

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STREAM PROCESSING

KAFKA SUPPORTS REAL-TIME PROCESSING OF STREAMS OF DATA USING TOOLS LIKE KAFKA STREAMS.

EXAMPLE: REAL-TIME ANALYTICS ON INCOMING TRANSACTION DATA.

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KAFKA CONNECT

KAFKA CONNECT SIMPLIFIES THE INTEGRATION OF KAFKA WITH OTHER DATA SYSTEMS.

EXAMPLE: USING KAFKA CONNECT TO SYNC DATA BETWEEN A DATABASE AND A KAFKA TOPIC.

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KAFKA STREAMS API

KAFKA STREAMS API ALLOWS BUILDING STREAM PROCESSING APPLICATIONS WITH KAFKA.

EXAMPLE: AN APPLICATION THAT AGGREGATES USER CLICKSTREAM DATA IN REAL-TIME.

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LOG COMPACTION

KAFKA CAN RETAIN ONLY THE LATEST MESSAGE PER KEY IN A LOG-COMPACTED TOPIC, USEFUL FOR CHANGELOG DATA.

EXAMPLE: KEEPING ONLY THE LATEST UPDATE TO USER PROFILES.

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EXACTLY ONCE SEMANTICS

KAFKA ENSURES THAT MESSAGES ARE PROCESSED EXACTLY ONCE, EVEN IN DISTRIBUTED SYSTEMS.

EXAMPLE: FINANCIAL TRANSACTIONS PROCESSED WITHOUT DUPLICATES.

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IDEMPOTENT PRODUCER

PRODUCERS CAN SAFELY RETRY SENDING MESSAGES WITHOUT DUPLICATING THEM.

EXAMPLE: SENDING A PAYMENT CONFIRMATION MESSAGE WITH GUARANTEED SINGLE DELIVERY.

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TRANSACTIONS

KAFKA SUPPORTS ATOMIC WRITES ACROSS MULTIPLE PARTITIONS AND TOPICS USING TRANSACTIONS.

EXAMPLE: ENSURING THAT A SERIES OF RELATED MESSAGES ARE EITHER ALL WRITTEN OR NONE ARE.

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MIRRORMAKER

TOOL FOR REPLICATING KAFKA TOPICS ACROSS CLUSTERS, USEFUL FOR DISASTER RECOVERY AND MULTI-DATACENTER SETUPS.

EXAMPLE: MIRRORING PRODUCTION DATA TO A BACKUP DATACENTER.

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SECURITY

KAFKA SUPPORTS AUTHENTICATION, AUTHORIZATION, AND ENCRYPTION TO SECURE DATA.

EXAMPLE: USING SSL FOR ENCRYPTING DATA IN TRANSIT.

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KAFKA ADMINCLIENT

ADMINCLIENT API ALLOWS PROGRAMMATIC MANAGEMENT OF KAFKA.

EXAMPLE: CREATING TOPICS, ALTERING CONFIGURATIONS PROGRAMMATICALLY.

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MONITORING AND METRICS

KAFKA PROVIDES METRICS AND MONITORING TOOLS TO TRACK CLUSTER PERFORMANCE.

EXAMPLE: MONITORING CONSUMER LAG AND BROKER HEALTH.

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SERIALIZATION AND DESERIALIZATION

KAFKA REQUIRES SERIALIZATION OF DATA FOR TRANSMISSION, WITH SUPPORT FOR VARIOUS FORMATS LIKE AVRO, JSON.

EXAMPLE: SERIALIZING USER DATA TO AVRO FORMAT BEFORE SENDING TO KAFKA.

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MESSAGE ORDERING

KAFKA MAINTAINS THE ORDER OF MESSAGES WITHIN A PARTITION, IMPORTANT FOR CONSISTENCY.

EXAMPLE: ENSURING ORDER OF TRANSACTION LOGS.

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CONSUMER GROUP

CONSUMERS CAN JOIN GROUPS TO BALANCE LOAD AND ENSURE EACH MESSAGE IS PROCESSED ONCE.

EXAMPLE: MULTIPLE CONSUMERS PROCESSING A HIGH-VOLUME TOPIC COLLABORATIVELY.

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OFFSET MANAGEMENT

KAFKA TRACKS THE OFFSET OF MESSAGES TO MANAGE CONSUMER PROGRESS.

EXAMPLE: STORING OFFSETS IN KAFKA TO RESUME PROCESSING AFTER A RESTART.

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TOPIC REPLICATION

KAFKA REPLICATES PARTITIONS ACROSS MULTIPLE BROKERS FOR FAULT TOLERANCE.

EXAMPLE: A PARTITION REPLICATED ACROSS THREE BROKERS TO HANDLE BROKER FAILURE.

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MESSAGE COMPRESSION

KAFKA SUPPORTS COMPRESSING MESSAGES TO SAVE BANDWIDTH AND STORAGE.

EXAMPLE: COMPRESSING LOG MESSAGES BEFORE SENDING TO KAFKA.

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ZOOKEEPER

KAFKA USES ZOOKEEPER FOR DISTRIBUTED COORDINATION AND METADATA MANAGEMENT.

EXAMPLE: ZOOKEEPER MANAGING BROKER METADATA AND LEADER ELECTION.

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KAFKA BROKER CONFIGURATION

BROKERS CAN BE CONFIGURED FOR PERFORMANCE, RETENTION POLICIES, AND MORE.

EXAMPLE: CONFIGURING A BROKER TO RETAIN MESSAGES FOR 30 DAYS.

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PRODUCER CONFIGURATION

PRODUCERS HAVE CONFIGURABLE PARAMETERS FOR MESSAGE DELIVERY, RETRIES, AND MORE.

EXAMPLE: SETTING PRODUCER RETRIES TO HANDLE TRANSIENT FAILURES.

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CONSUMER CONFIGURATION

CONSUMERS CAN BE CONFIGURED FOR FETCH SIZES, TIMEOUT SETTINGS, AND MORE.

EXAMPLE: CONFIGURING CONSUMER FETCH SIZE FOR OPTIMAL PERFORMANCE.

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TOPIC MANAGEMENT

TOPICS CAN BE CREATED, DELETED, AND MANAGED PROGRAMMATICALLY OR VIA CLI.

EXAMPLE: CREATING A NEW TOPIC FOR STORING EVENT LOGS.

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QUOTAS AND THROTTLING

KAFKA SUPPORTS SETTING QUOTAS TO CONTROL RESOURCE USAGE BY CLIENTS.

EXAMPLE: THROTTLING A HIGH-VOLUME PRODUCER TO PREVENT OVERWHELMING THE CLUSTER.

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REBALANCE PROTOCOL

KAFKA HANDLES REBALANCING OF CONSUMERS WITHIN A GROUP TO MAINTAIN LOAD BALANCE.

EXAMPLE: REBALANCING PARTITIONS WHEN A NEW CONSUMER JOINS THE GROUP.

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KAFKA REST PROXY

PROVIDES A RESTFUL INTERFACE TO INTERACT WITH KAFKA CLUSTERS.

EXAMPLE: SENDING MESSAGES TO KAFKA USING HTTP REQUESTS.

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KAFKA API

KAFKA PROVIDES APIS FOR PRODUCING, CONSUMING, AND MANAGING DATA.

EXAMPLE: USING THE KAFKA PRODUCER API TO SEND MESSAGES FROM A JAVA APPLICATION.

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SCHEMA REGISTRY

CONFLUENT SCHEMA REGISTRY MANAGES AND ENFORCES SCHEMAS FOR KAFKA MESSAGES.

EXAMPLE: ENSURING ALL MESSAGES IN A TOPIC FOLLOW A PREDEFINED SCHEMA.

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KAFKA STREAMS DSL

A HIGH-LEVEL API FOR STREAM PROCESSING IN KAFKA.

EXAMPLE: USING KAFKA STREAMS DSL TO FILTER AND TRANSFORM A STREAM OF EVENTS.

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FAULT TOLERANCE

KAFKA'S DESIGN ENSURES HIGH AVAILABILITY AND FAULT TOLERANCE.

EXAMPLE: AUTOMATIC FAILOVER TO REPLICAS WHEN A BROKER FAILS.

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REAL-TIME ANALYTICS

KAFKA SUPPORTS REAL-TIME DATA ANALYTICS AND PROCESSING.

EXAMPLE: REAL-TIME DASHBOARD UPDATING WITH LIVE METRICS FROM KAFKA.

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ETL PIPELINES

KAFKA CAN BE USED TO BUILD EFFICIENT ETL PIPELINES FOR DATA INTEGRATION.

EXAMPLE: EXTRACTING DATA FROM DATABASES, TRANSFORMING IT, AND LOADING IT INTO A DATA WAREHOUSE VIA KAFKA.

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KAFKA UPGRADES

KAFKA SUPPORTS ROLLING UPGRADES TO MINIMIZE DOWNTIME.

EXAMPLE: UPGRADING KAFKA BROKERS WITHOUT DISRUPTING MESSAGE FLOW.

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MESSAGE TIMESTAMPING

KAFKA MESSAGES CAN HAVE TIMESTAMPS FOR TIME-BASED PROCESSING.

EXAMPLE: USING TIMESTAMPS FOR EVENT TIME PROCESSING IN KAFKA STREAMS.

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STATE STORES

KAFKA STREAMS ALLOWS MAINTAINING STATEFUL PROCESSING WITH STATE STORES.

EXAMPLE: COUNTING OCCURRENCES OF EVENTS OVER A WINDOW OF TIME USING STATE STORES.

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WINDOWED OPERATIONS

KAFKA STREAMS SUPPORTS WINDOWED OPERATIONS FOR AGGREGATIONS OVER TIME WINDOWS.

EXAMPLE: CALCULATING THE SUM OF TRANSACTIONS EVERY MINUTE.

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KSQL

KSQL IS A SQL-LIKE INTERFACE FOR STREAM PROCESSING WITH KAFKA.

EXAMPLE: USING KSQL TO PERFORM REAL-TIME FILTERING AND AGGREGATIONS ON KAFKA TOPICS.

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KAFKA ECOSYSTEM

KAFKA'S ECOSYSTEM INCLUDES TOOLS LIKE CONNECT, STREAMS, KSQL, AND MORE FOR COMPREHENSIVE DATA PROCESSING.

EXAMPLE: USING KAFKA CONNECT TO INTEGRATE WITH DATABASES, KAFKA STREAMS FOR PROCESSING, AND KSQL FOR QUERYING STREAMS.

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KAFKA CONNECTORS

PRE-BUILT CONNECTORS FOR INTEGRATING KAFKA WITH VARIOUS DATA SOURCES AND SINKS.

EXAMPLE: USING A JDBC CONNECTOR TO SYNC DATA BETWEEN A DATABASE AND KAFKA.

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KAFKA CLUSTER MANAGEMENT

TOOLS AND PRACTICES FOR MANAGING KAFKA CLUSTERS EFFICIENTLY.

EXAMPLE: USING TOOLS LIKE KAFKA MANAGER FOR MONITORING AND MANAGING CLUSTER HEALTH.

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TIERED STORAGE

KAFKA'S TIERED STORAGE ALLOWS OFFLOADING OLDER DATA TO CHEAPER STORAGE.

EXAMPLE: STORING OLDER KAFKA TOPIC DATA IN S3 TO REDUCE ON-PREM STORAGE COSTS.

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